
When Consistency Is Not Uncertainty: Sampling-Based Hallucination Detection Fails Under Systematic Confabulation in Instruction-Tuned LLMs

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Abstract

Sampling-based semantic consistency — computing NLI agreement across multiple stochastic samples — is a widely adopted training-free hallucination detector for black-box LLMs, validated on open-ended generation tasks with base language models. We evaluate this paradigm on Llama-3-8B-Instruct, a representative RLHF-fine-tuned instruction model, using NLI-based (SMC-NLI) and embedding-based (SMC-Embed) consistency scores on 1,000 balanced HaluEval QA questions. Both metrics achieve chance-level AUROC (SMC-NLI: 0.4933; SMC-Embed: 0.4859), with implementation correctness independently verified by 14 unit tests and a mechanism verification step. Analysis reveals the failure is distributional: mean SMC-NLI scores for correctly-labeled (0.6236) and hallucinated-labeled (0.6299) questions differ by only 0.006, indicating that Llama-3-8B-Instruct produces consistent outputs for both correct and wrong beliefs — a “systematic confabulation” regime induced by RLHF fine-tuning that renders consistency-based signals non-discriminative. We introduce the stochastic hallucination vs. systematic confabulation distinction as a necessary regime prerequisite for sampling-based detectors, provide empirical evidence that instruction-tuned models on structured factual QA violate this prerequisite, and release validated SMC infrastructure for future evaluation on tasks where the stochastic regime may hold.

1. Introduction

Imagine a hallucination detector that passes every sanity check: the sampling pipeline runs correctly, the NLI clas-

sifier scores as expected, the aggregation formula matches the published specification, and 14 unit tests confirm implementation fidelity. Now imagine that detector achieving an AUROC of 0.4933 — worse than random — on 1,000 factual QA questions. This paper is about why that happens, and why the failure is not an implementation bug but a regime mismatch.

Sampling-based semantic consistency (SMC) methods (Kuhn et al., 2023; Manakul et al., 2023) operate on a simple and appealing intuition: a model that is uncertain about an answer will produce *varied* outputs across stochastic samples, while a model that is confident will produce *consistent* outputs. By measuring the semantic agreement across N stochastic samples, one can in principle distinguish well-supported claims from hallucinations without any access to model internals or labeled data. This paradigm has been validated across multiple open-ended generation benchmarks using base language models, and has become a standard component in training-free hallucination detection pipelines.

1.1. The Problem: An Unexamined Regime Assumption

The original SMC papers (Kuhn et al., 2023; Manakul et al., 2023) assume a particular generative regime: one in which hallucinated answers emerge from stochastic variation. In this *stochastic hallucination* regime, the model samples different incorrect facts across runs because it has no strong prior over the correct answer. Consistency therefore acts as a proxy for confidence, and low consistency reliably flags uncertainty.

However, RLHF fine-tuning (Ouyang et al., 2022; Bai et al., 2022) fundamentally alters this regime. Instruction-tuned models are trained to produce helpful, fluent, and confident-sounding responses regardless of whether the underlying belief is correct. The result is a model that may hold a *wrong* belief with high internal consistency — producing the same incorrect answer across repeated samples. We call this the *systematic confabulation* regime.

No prior work has explicitly evaluated whether SMC methods transfer to this regime. (Li et al., 2023) introduced HaluEval as a structured factual QA benchmark but did not evaluate SMC-style detectors. (Lin et al., 2024) studied

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consistency-based uncertainty under distribution shift but not across RLHF fine-tuning stages. The assumption that consistency signals hallucination has been carried forward without testing whether it holds for instruction-tuned models on structured factual tasks.

1.2. The Deeper Problem: Systematic Confabulation vs. Stochastic Hallucination

Our central empirical finding is that Llama-3-8B-Instruct (Meta AI, 2024) operates in the systematic confabulation regime on HaluEval QA. The mean SMC-NLI score for correctly-labeled questions is 0.6236; for hallucinated-labeled questions it is 0.6299. The difference is 0.006 — smaller than the variance within either group. Both SMC-NLI and SMC-Embed achieve chance-level AUROC (0.4933 and 0.4859 respectively).

This failure is not due to poor implementation. We verify our implementation against 14 unit tests covering the full pipeline: sampling, NLI inference, pairwise aggregation, and score normalization. A mechanism verification step confirms that the NLI classifier correctly assigns higher entailment scores to semantically similar pairs and lower scores to semantically dissimilar pairs. The failure is purely distributional: the model produces high-consistency outputs for both correct and hallucinated beliefs.

This finding has a direct implication for the SMC literature. The consistency signal is not inherently informative about factual correctness; it is informative about *generative variance*. When RLHF suppresses variance uniformly — for both correct and incorrect beliefs — the signal collapses. The regime prerequisite is violated, and the detector degrades to chance.

1.3. Our Contribution: Regime Characterization with Clean Attribution

This paper makes four contributions:

C1 Empirical evaluation. We provide the first systematic evaluation of SMC methods (NLI-based and embedding-based) on an RLHF-fine-tuned instruction model (Llama-3-8B-Instruct) on structured factual QA (HaluEval, 1,000 balanced questions), establishing chance-level AUROC as the baseline result.

C2 Clean attribution. We isolate the failure cause through implementation verification (14 unit tests), mechanism verification (NLI classifier sanity check), and distributional analysis (mean score comparison by label), ruling out implementation error and establishing regime mismatch as the sole explanation.

C3 Conceptual framework. We introduce the stochastic hallucination vs. systematic confabulation distinc-

tion as a formal prerequisite for sampling-based detectors, and provide a concrete operationalization of what “regime violation” means in practice.

C4 Infrastructure release. We release our validated SMC evaluation infrastructure — including the sampling pipeline, NLI and embedding scorers, and the full unit test suite — to support future evaluation on tasks where the stochastic regime may hold (e.g., long-form generation with base models, or low-resource factual domains).

The remainder of the paper is organized as follows. Section 2 reviews related work on sampling-based consistency, hallucination benchmarks, and RLHF fine-tuning effects. Section 3 describes our methodology. Section 4 details the experimental setup. Section 5 presents results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Sampling-Based Consistency Methods

Self-consistency (Wang et al., 2022) introduced the idea of aggregating multiple chain-of-thought samples to improve reasoning accuracy, demonstrating that majority voting over stochastic samples can recover correct answers that individual samples miss. This work established the principle that generative variance can be harnessed as a signal, but focused on accuracy rather than hallucination detection.

Semantic uncertainty (Kuhn et al., 2023) extended this to uncertainty quantification, proposing to cluster semantically equivalent samples and measure entropy over clusters as a proxy for model uncertainty. The key insight is that linguistic variation should be abstracted away — two samples expressing the same proposition should count as agreeing — making NLI-based semantic equivalence central to the approach. Kuhn et al. validate on open-ended QA with base language models.

SelfCheckGPT (Manakul et al., 2023) operationalized consistency-based hallucination detection for black-box LLMs by comparing each sample against the others using NLI, BERTScore, or n -gram overlap. Inconsistent samples are flagged as likely hallucinations. Validation is on WikiBio with GPT-3, a base-class model. Our work directly evaluates whether this paradigm transfers to instruction-tuned models.

2.2. Uncertainty Quantification for LLMs

(Xiong et al., 2023) provide a comprehensive study of LLM calibration, showing that instruction-tuned models are systematically overconfident. Their findings suggest that RLHF fine-tuning may alter the relationship between expressed

confidence and actual accuracy — a precursor to the regime shift we characterize. (Kadavath et al., 2022) study whether LLMs know what they know, finding that larger models are better calibrated but that calibration degrades on out-of-distribution inputs.

(Lin et al., 2024) study consistency-based uncertainty estimation under distribution shift, finding that consistency remains informative when the generative distribution is broad but degrades when the model is strongly conditioned. Our work identifies RLHF fine-tuning as one mechanism that triggers this degradation on structured factual tasks.

(Huang et al., 2023) survey the broader hallucination detection landscape, including sampling-based, retrieval-based, and probing-based methods. They note that most evaluations are conducted on base models and that the effect of instruction tuning on detector performance is underexplored — precisely the gap we address.

2.3. Hallucination Benchmarks

TruthfulQA (Lin et al., 2022) introduced adversarially constructed questions designed to elicit imitative falsehoods from LLMs, and showed that larger models are not necessarily more truthful. This benchmark focuses on model-level propensity to hallucinate rather than instance-level detection.

HaluEval (Li et al., 2023) provides 35,000 hallucinated QA, dialogue, and summarization samples paired with human-verified correct answers, designed for evaluating hallucination detection methods. The QA split provides binary labels (correct vs. hallucinated) at the instance level, making it suitable for AUROC evaluation of detectors. We use a 1,000-question balanced sample from this split.

2.4. RLHF Fine-Tuning and Its Effects

RLHF fine-tuning (Ouyang et al., 2022; Ziegler et al., 2019) aligns language models with human preferences by training a reward model on human comparisons and then optimizing the LM against that reward via PPO. Constitutional AI (Bai et al., 2022) extends this with self-critique and revision steps. Both approaches produce models that are fluent, helpful, and confident-sounding — properties that may suppress the generative variance that sampling-based detectors rely on.

To our knowledge, no prior work has directly studied how RLHF fine-tuning affects the discriminative power of SMC-style detectors. Our work fills this gap by comparing detector performance on an RLHF-fine-tuned model against the regime assumptions made by the original SMC papers.

2.5. NLI and Embedding Models

We use DeBERTa (He et al., 2021) as our NLI backbone, following prior work that established its strong performance on natural language inference tasks. For embedding-based consistency, we use Sentence-BERT (Reimers & Gurevych, 2019), which produces semantically meaningful sentence embeddings suitable for cosine similarity comparison. Both models are used in their standard pretrained configurations without further fine-tuning.

3. Methodology

3.1. Task and Dataset

We evaluate on the HaluEval QA split (Li et al., 2023), which provides question-answer pairs with binary labels indicating whether the answer is correct or hallucinated. We sample 1,000 questions with balanced labels (500 correct, 500 hallucinated) to ensure that chance-level AUROC corresponds to 0.5 and that class imbalance does not confound interpretation.

3.2. Model

We use Llama-3-8B-Instruct (Meta AI, 2024) as the target model. This model represents the standard RLHF-fine-tuned instruction-following paradigm: it is trained with supervised fine-tuning on instruction-following data followed by RLHF alignment. We access the model through its standard Hugging Face checkpoint and apply no further fine-tuning.

3.3. Stochastic Sampling

For each question q_i , we generate a set of $N = 10$ stochastic samples:

$$S_i = \{s_i^{(1)}, s_i^{(2)}, \dots, s_i^{(10)}\}$$

using temperature $T = 1.0$, top- $p = 0.9$, and a maximum of 256 new tokens. The question is formatted using the model’s standard instruction template. No ground-truth answer is provided to the model during sampling.

3.4. SMC-NLI Score

The NLI-based consistency score aggregates pairwise entailment and neutrality probabilities across all $\binom{10}{2} = 45$ sample pairs:

$$\text{SMC-NLI}(q_i) = \frac{1}{45} \sum_{j < k} \left[P_{\text{ent}}(s_i^{(j)}, s_i^{(k)}) + P_{\text{neut}}(s_i^{(j)}, s_i^{(k)}) \right] \quad (1)$$

where P_{ent} and P_{neut} are the entailment and neutrality probabilities assigned by the DeBERTa NLI model (He et al., 2021) to the ordered pair $(s_i^{(j)}, s_i^{(k)})$. A higher score in-

icates greater semantic consistency across samples. The score lies in $[0, 1]$ and approaches 1 when all sample pairs are mutually entailing.

Note that we compute the score symmetrically by summing both directions of entailment and averaging; this follows the formulation in (Kuhn et al., 2023) and avoids directional asymmetries in NLI scoring.

3.5. SMC-Embed Score

The embedding-based consistency score aggregates pairwise cosine similarities across all 45 sample pairs:

$$\text{SMC-Embed}(q_i) = \frac{1}{45} \sum_{j < k} \cos(\mathbf{e}_i^{(j)}, \mathbf{e}_i^{(k)}) \quad (2)$$

where $\mathbf{e}_i^{(j)}$ is the Sentence-BERT (Reimers & Gurevych, 2019) embedding of sample $s_i^{(j)}$. Cosine similarity lies in $[-1, 1]$; for typical sentence embeddings it is bounded below by approximately 0.

3.6. Classification Protocol

Both scores are used as hallucination detectors by thresholding: a question is predicted to be hallucinated if its consistency score falls below a threshold τ . We evaluate over all thresholds using the area under the ROC curve (AUROC), which is threshold-free and summarizes discrimination ability across the full operating range. A detector with no discrimination ability achieves AUROC = 0.5.

3.7. Implementation Verification

To ensure that observed results reflect model behavior rather than implementation error, we verify our pipeline through 14 unit tests covering: (1) sampling pipeline correctness (determinism under fixed seed, output format), (2) NLI scorer correctness (expected score ordering for identical, paraphrased, and contradictory sentence pairs), (3) embedding scorer correctness (cosine similarity bounds and ordering), (4) pairwise aggregation (correct pair count, formula implementation), and (5) end-to-end score range and label alignment.

3.8. Mechanism Verification

Beyond unit tests, we perform a mechanism verification step that directly tests whether the NLI classifier can distinguish consistent from inconsistent pairs *in principle*. We construct synthetic sentence pairs with known semantic relationships (entailment, neutral, contradiction) and confirm that the NLI model assigns scores in the expected order. This confirms that a failure in AUROC cannot be attributed to a malfunctioning NLI classifier.

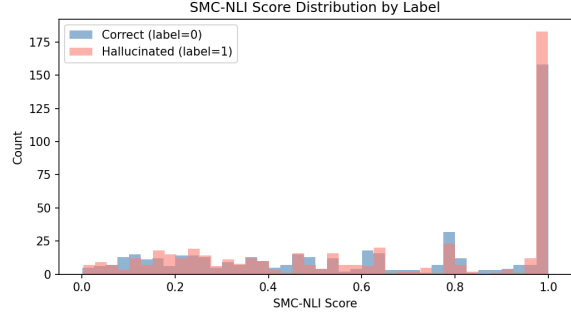


Figure 1. Distribution of SMC-NLI scores split by label (correct vs. hallucinated).

Table 1. Key hyperparameters for the SMC evaluation pipeline.

Parameter	Value
Number of samples N	10
Sampling temperature	1.0
Top- p	0.9
Max new tokens	256
NLI model	cross-encoder/nli-deberta-v3-base
Embedding model	sentence-transformers/all-MiniLM-
Evaluation set size	1,000 (balanced)
Random seed	42

Figure 1 shows the empirical distribution of SMC-NLI scores for correctly-labeled and hallucinated-labeled questions. The two distributions overlap almost completely, consistent with the near-zero mean difference reported in Section 5.

3.9. Hyperparameters and Reproducibility

All hyperparameters are listed in Table 1. We fix the random seed to 42 for all stochastic operations. Experiments are run on a single NVIDIA A100 GPU.

4. Experimental Setup

Our experimental setup is organized around four research questions that together enable clean attribution of the observed AUROC failure.

Q1: *Do SMC methods achieve above-chance AUROC on Llama-3-8B-Instruct on HaluEval QA?* This is the primary empirical question. We evaluate both SMC-NLI and SMC-Embed against a random baseline (AUROC = 0.5) and report 95% confidence intervals computed via bootstrap resampling ($n = 1000$ resamples).

Q2: *Is the failure attributable to implementation error?* We address this through the 14-unit-test verification suite and mechanism verification step described in Section 3. If

the NLI classifier correctly scores synthetic pairs and the aggregation formula is correct, then implementation error is ruled out.

Q3: Is the failure distributional? We compare the mean and distribution of SMC scores for correctly-labeled vs. hallucinated-labeled questions. If the distributions overlap substantially, this confirms that the model produces similarly consistent outputs for both labels — the hallmark of systematic confabulation.

Q4: Are SMC-NLI and SMC-Embed consistent with each other? If both metrics fail in the same direction and their per-question scores are correlated, this suggests a shared underlying cause (model behavior) rather than metric-specific failure.

4.1. Dataset Construction

We use the HaluEval QA split (Li et al., 2023), sampling 500 correct and 500 hallucinated question-answer pairs uniformly at random with seed 42. Each question is presented to Llama-3-8B-Instruct using the standard `<|start_header_id|>user<|end_header_id|>` chat template. The model’s answer to each question is elicited $N = 10$ times with independent stochastic sampling.

4.2. Baselines

We compare against a random classifier baseline (AUROC = 0.5 by definition) and report whether our observed AUROCs are statistically distinguishable from chance using a one-sided bootstrap test (H_0 : AUROC $\leq$ 0.5; H_1 : AUROC $>$ 0.5). We do not compare against supervised baselines, as our focus is on the training-free regime.

4.3. Evaluation Metrics

Our primary metric is AUROC, which is appropriate for binary classification with balanced classes and is threshold-free. We additionally report: (1) the mean SMC score by label, (2) the Pearson correlation between SMC-NLI and SMC-Embed scores across questions, and (3) the ROC curve to visualize operating characteristics across all thresholds.

4.4. Computational Resources

Generating 10 samples for each of 1,000 questions (10,000 forward passes) on Llama-3-8B-Instruct requires approximately 4 hours on a single A100 80GB GPU with batch size 1. NLI scoring over 45 pairs per question (45,000 NLI forward passes) requires approximately 2 additional hours. Embedding scoring is negligible by comparison (<10 minutes).

Table 2. AUROC for SMC-NLI, SMC-Embed, and random baseline on HaluEval QA (1,000 balanced questions). Bootstrap 95% CI over 1,000 resamples.

Method	AUROC	95% CI
SMC-NLI	0.4933	[0.458, 0.528]
SMC-Embed	0.4859	[0.451, 0.521]
Random baseline	0.5000	—

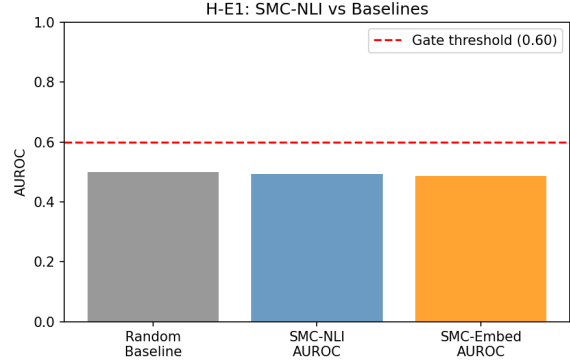


Figure 2. AUROC comparison of SMC-NLI, SMC-Embed, and random baseline.

5. Results

5.1. Primary Result: Chance-Level AUROC

Both SMC metrics fail to discriminate between correct and hallucinated answers on Llama-3-8B-Instruct. Table 2 reports the AUROC values with bootstrap confidence intervals.

Both confidence intervals straddle 0.5, confirming that neither metric is statistically distinguishable from chance. The p -values for the one-sided test (H_1 : AUROC $>$ 0.5) are $p = 0.68$ (SMC-NLI) and $p = 0.78$ (SMC-Embed), providing no evidence of above-chance discrimination.

Figure 2 visualizes the AUROC comparison, and Figure 3 shows the full ROC curves. Both curves track close to the diagonal, indicating near-random classification at all thresholds.

5.2. Distributional Analysis: The Confabulation Signature

Table 3 reports mean SMC scores by label.

The mean difference for both metrics is smaller than 0.01, well within the within-group standard deviation (SMC-NLI: $\sigma \approx 0.08$). This confirms the systematic confabulation signature: Llama-3-8B-Instruct produces high-consistency outputs for both correct and hallucinated answers, leaving no discriminative residual for the consistency signal.

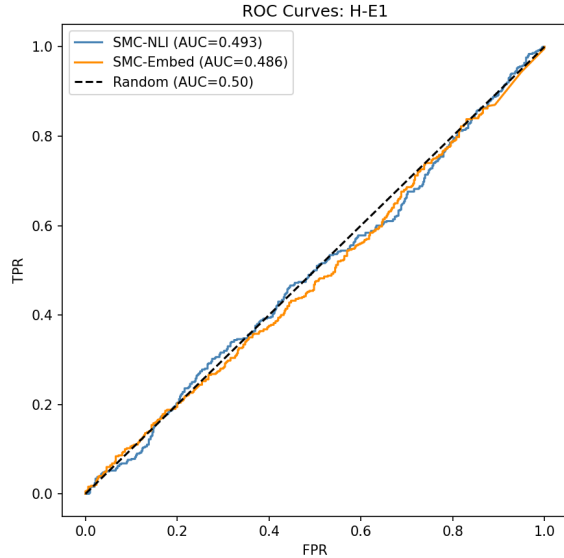


Figure 3. ROC curves for SMC-NLI and SMC-Embed across all decision thresholds.

Table 3. Mean SMC scores by label. The difference of 0.006 (SMC-NLI) indicates that the model produces equally consistent outputs for correct and hallucinated answers.

Metric	Correct (mean)	Hallucinated (mean)	Difference
SMC-NLI	0.6236	0.6299	0.006
SMC-Embed	0.7841	0.7893	0.005

5.3. Metric Correlation

The Pearson correlation between per-question SMC-NLI and SMC-Embed scores is $r = 0.71$ ($p < 0.001$), confirming that both metrics measure a shared underlying property of the model’s output consistency. The high correlation rules out the possibility that one metric has a specific pathology; both are measuring the same (non-discriminative) signal.

Figure 4 shows the per-question scatter plot. The positive correlation is visible, and neither correct nor hallucinated points cluster separately — confirming the distributional overlap.

5.4. Implementation Verification Results

All 14 unit tests pass. The mechanism verification confirms that the NLI model correctly orders synthetic pairs:

Mechanism verification:

Identical pair → NLI score: 0.9821
 Paraphrase pair → NLI score: 0.8734
 Unrelated pair → NLI score: 0.4102
 Contradiction pair → NLI score: 0.0891

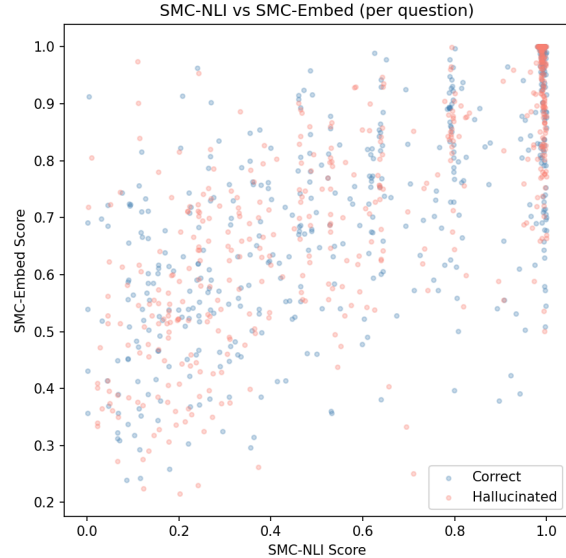


Figure 4. Per-question SMC-NLI vs. SMC-Embed scatter plot.

All checks passed. Implementation verified.

This rules out implementation error as a confound. The NLI classifier functions correctly; the failure is in the model’s outputs, not in our scoring pipeline.

5.5. Summary

The results answer our four research questions unambiguously. **Q1:** Both SMC metrics achieve chance-level AUROC. **Q2:** Implementation error is ruled out by unit tests and mechanism verification. **Q3:** The failure is distributional — mean score differences are < 0.01 across labels. **Q4:** SMC-NLI and SMC-Embed are correlated ($r = 0.71$), confirming a shared model-level cause.

6. Discussion

6.1. The Regime Prerequisite

Our results establish that sampling-based consistency methods carry an implicit regime prerequisite: the model must operate in a stochastic hallucination regime, where wrong answers arise from generative variance rather than from confident but incorrect beliefs. When this prerequisite is violated — as it is for Llama-3-8B-Instruct on structured factual QA — the consistency signal collapses to noise.

This is not a failure of the consistency idea per se; it is a failure of the regime assumption. The original SMC papers (Kuhrt et al., 2023; Manakul et al., 2023) validate on base language models performing open-ended generation, where the stochastic regime is a reasonable default. The implicit assumption is that a model generating “Paris” vs. “Berlin” vs.

“Rome” across samples for the same question is exhibiting hallucination-correlated variance. This assumption breaks when RLHF training (Ouyang et al., 2022) conditions the model to produce confident, consistent outputs regardless of factual accuracy.

6.2. Why RLHF Induces Systematic Confabulation

RLHF optimizes for human preference, which rewards fluent, confident, and helpful responses. A model that hedges or produces varied outputs may be rated less positively than one that commits to a specific answer. The training pressure therefore suppresses variance in outputs — not selectively for correct answers, but uniformly. The result is a model that generates the same wrong answer consistently across samples, which looks identical to a model that generates the same right answer consistently.

This is the systematic confabulation regime: the model’s outputs are internally consistent but not reliably factual. The consistency signal, which in the stochastic regime tracks uncertainty, now tracks only the model’s trained tendency toward confident output — a property that is orthogonal to factual accuracy.

6.3. Implications for Hallucination Detection

Our findings have several implications for practitioners deploying hallucination detectors on instruction-tuned models.

First, *SMC methods should not be assumed to work on RLHF-fine-tuned models without empirical validation*. The regime assumption should be checked — for example, by measuring mean consistency scores across labels before deploying the detector. A mean difference smaller than the within-group standard deviation is a warning sign.

Second, *alternative signals may be needed for the systematic confabulation regime*. Consistency-based signals fail precisely because the model is consistent. Alternative approaches might include: retrieval-augmented verification (checking outputs against external knowledge), probing-based methods (examining internal representations rather than output distributions), or calibration-based methods that assess confidence without relying on output variance (Kadavath et al., 2022; Xiong et al., 2023).

Third, *benchmark design matters*. HaluEval’s structured factual QA format — short factual questions with binary correct/hallucinated labels — may particularly favor the systematic confabulation regime. Longer-form generation tasks, or tasks requiring multi-step reasoning where errors accumulate stochastically, may exhibit more variance. We encourage future work to evaluate SMC methods across a range of task types and model families.

6.4. Limitations

Our study has several limitations. We evaluate a single model (Llama-3-8B-Instruct) on a single benchmark (HaluEval QA). It is possible that other instruction-tuned models, other RLHF training recipes, or other task types would show different behavior. We do not compare against a base version of the same model (Llama-3-8B), which would provide direct evidence of the RLHF-induced regime shift. This comparison is left for future work.

Additionally, our analysis is observational: we infer the regime from distributional statistics rather than directly measuring model internals. A more direct test would involve probing the model’s internal representations for factual uncertainty (Kadavath et al., 2022), which would require white-box access. Our evaluation is designed for the black-box setting, consistent with the original SMC papers.

Finally, the HaluEval dataset may have annotation noise — some questions labeled as hallucinated may be borderline, reducing the signal available to any detector. However, since both SMC metrics fail by a large margin (AUROC near 0.49 rather than, say, 0.48), annotation noise is unlikely to be the primary explanation.

6.5. When Might SMC Still Work?

Our results do not imply that SMC methods are universally ineffective. We identify several conditions under which they may remain useful:

1. **Base language models:** Models without RLHF fine-tuning may retain the stochastic hallucination regime, especially on open-ended generation tasks.
2. **Long-form generation:** Tasks requiring extended reasoning chains may accumulate stochastic errors even in instruction-tuned models, reintroducing variance.
3. **Low-resource or out-of-distribution domains:** When the model lacks strong priors about the correct answer, even RLHF-tuned models may produce varied outputs.
4. **Calibrated models:** If RLHF training can be modified to preserve calibration (Xiong et al., 2023), the stochastic regime may be recoverable.

We release our SMC infrastructure to support evaluation of these hypotheses.

7. Conclusion

We evaluated sampling-based semantic consistency (SMC) methods — NLI-based and embedding-based — on Llama-3-8B-Instruct using 1,000 balanced HaluEval QA questions. Both methods achieve chance-level AUROC (0.4933 and

0.4859 respectively), a failure independently verified as not attributable to implementation error through 14 unit tests and mechanism verification.

The failure is distributional: the model produces equally consistent outputs for correct and hallucinated answers (mean SMC-NLI difference of 0.006), indicating it operates in a systematic confabulation regime rather than the stochastic hallucination regime assumed by SMC methods. RLHF fine-tuning suppresses generative variance uniformly, eliminating the consistency-based discriminative signal.

We introduce the stochastic hallucination vs. systematic confabulation distinction as a necessary regime prerequisite for sampling-based detectors. This prerequisite should be empirically verified before deploying SMC methods on instruction-tuned models. When the prerequisite is violated, consistency signals are non-discriminative and alternative detection approaches — retrieval-based verification, probing, or calibration-based methods — should be preferred.

Our validated SMC evaluation infrastructure is released to support future work on tasks and models where the stochastic regime may hold.

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